

Southeast University Dhaka

Department of Electrical and Electronic Engineering



Design and implementation of an IoT-based smart data center cooling and environmental monitoring system

Submitted by

- | | |
|-----------------------------|-------------------|
| 1. MD Alim | ID- 2022200520074 |
| 2. Shakibur Rahman | ID- 2022200520015 |
| 3. Sinthiya Jahan | ID- 2022200520086 |
| 4. MD. Nahidur Rahman Akash | ID- 2022200520027 |

Supervised by

Mr. Md. Touhidul Islam

Lecturer

Department of Electrical and Electronic Engineering
Southeast University Dhaka

Declaration of Candidates

We, the undersigned, hereby declare that the project work entitled "**Design and Implementation of an IoT-Based Smart Data Center Cooling and Environmental Monitoring System**" is our original work. This project was carried out by us under the supervision of **Mr. Md. Touhidul Islam**, Lecturer, Department of Electrical and Electronic Engineering (EEE), Southeast University, Dhaka.

We further declare that this project, or any part of it, has not been submitted elsewhere for the award of any degree or diploma. All the information and data presented in this report have been gathered through our own research, design, and hardware implementation

Candidates' Details:

MD Alim
ID- 2022200520074

Shakibur Rahman
ID- 2022200520015

Sinthiya Jahan
ID- 2022200520086

MD. Nahidur Rahman Akash
ID- 2022200520027

Certificate

This is to certify that the project report entitled "**Design and Implementation of an IoT-Based Smart Data Center Cooling and Environmental Monitoring System**" is a record of the work carried out by **MD Alim (ID: 2022200520074)**, **Shakibur Rahman (ID: 2022200520015)**, **Sinhiya Jahan (ID: 2022200520086)**, and **MD. Nahidur Rahman Akash (ID: 2022200520027)** under my direct supervision and guidance.

To the best of my knowledge, the work presented in this report is original and meets the academic requirements for the completion of their degree in Electrical and Electronic Engineering (EEE).

I wish them every success in their future professional and academic endeavours

Supervisor,

Mr. Md. Touhidul Islam

Lecturer

Department of Electrical and Electronic Engineering

Southeast University Bangladesh

Dhaka, Bangladesh

Board of Examiners

Mr. Md. Touhidul Islam

(Supervisor)

Lecturer, Dept. of EEE

Southeast University Bangladesh (SUB)

1. Mohammad Shafiul Alam

(Examiner)

Assistant Professor, Dept. of EEE

Southeast University Bangladesh (NUB)

2. Sayemul Islam

(Examiner)

Lecturer, Dept. of EEE

Southeast University Bangladesh (NUB)

3. Md. Nazmul Islam

(Examiner)

Lecturer, Dept. of EEE

Southeast University Bangladesh (NUB)

DR. MD. MIZANUR RAHMAN

Associate Professor & Head, Dept. of EEE

Southeast University Bangladesh (NUB)

Abstract

Data centers are highly sensitive environments where temperature, humidity, air quality, and safety conditions must be maintained within strict limits to ensure uninterrupted operation of critical IT infrastructure. Inadequate cooling and lack of real-time environmental monitoring can lead to equipment failure, energy inefficiency, and increased operational risk.

This thesis presents the **design and implementation of an IoT-based smart data center cooling and environmental monitoring system**. The proposed system continuously monitors key environmental parameters such as temperature, humidity, smoke concentration, air quality index, equivalent carbon dioxide (eCO₂), and total volatile organic compounds (TVOC) using MQ-2, ENS160, and AHT21 sensors. An ESP32 microcontroller is used as the core processing unit, providing real-time data acquisition, decision-making, and wireless communication.

The system implements an **automatic dual air-conditioner (AC) control strategy** based on configurable high and low temperature setpoints, which are stored in non-volatile EEPROM memory to ensure data retention during power failure. Manual override of both AC units is enabled through physical push buttons as well as remote control via a Telegram bot. Environmental data and system status are transmitted to the Telegram platform at three-minute intervals, allowing remote monitoring and control.

In addition, the system incorporates a multi-level alarm and indication mechanism using LEDs and a buzzer to alert operators during abnormal conditions such as high temperature, improper humidity levels, poor air quality, or smoke detection. Experimental results demonstrate that the proposed system improves cooling efficiency, enhances operational safety, and provides a reliable, low-cost, and scalable solution for modern data center environments.

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First and foremost, we express our profound gratitude to the **Almighty**, whose blessings provided us with the strength and perseverance to complete this project successfully.

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The Candidates

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Chapter 1 — Introduction

1.1 Background of the Study

The rapid growth of cloud computing, digital services, and networked systems has increased the dependence on data centers as a critical part of modern infrastructure. Data centers support servers, networking equipment, and storage systems that operate continuously and generate significant heat during operation. Because of this, maintaining stable environmental conditions inside server rooms is essential for ensuring reliable performance, preventing hardware damage, and reducing unexpected downtime.

Temperature is one of the most important parameters in data center management. Excessive heat can reduce equipment lifespan, lower system efficiency, and increase the risk of system failure. At the same time, improper humidity levels can create additional risks, such as electrostatic discharge or moisture-related damage. Environmental concerns also extend beyond temperature and humidity. Smoke, harmful gases, and poor air quality can threaten both electronic equipment and personnel safety.

Traditional cooling systems in many small and medium-sized server rooms still rely on fixed air-conditioning schedules or manual supervision. These methods often lack real-time monitoring, adaptive control, and rapid response to changing conditions. As a result, they may lead to inefficient cooling, increased energy use, and delayed reaction during abnormal environmental events.

The advancement of the Internet of Things (IoT) has created opportunities to improve monitoring and control through connected sensors, microcontrollers, and wireless communication. IoT-based systems can collect environmental data in real time, analyze conditions, and automatically respond through intelligent control logic. These capabilities make IoT a practical approach for improving cooling efficiency, safety, and operational reliability in data center environments.

This project presents the design and implementation of an IoT-based data center cooling, environmental control, and alert data logging system. The proposed system combines environmental sensing, automatic dual air-conditioner control, alarm mechanisms, and remote monitoring through a Telegram bot within a unified framework. The goal is to provide a practical, scalable, and low-cost solution for improving environmental management in data centers and server rooms.

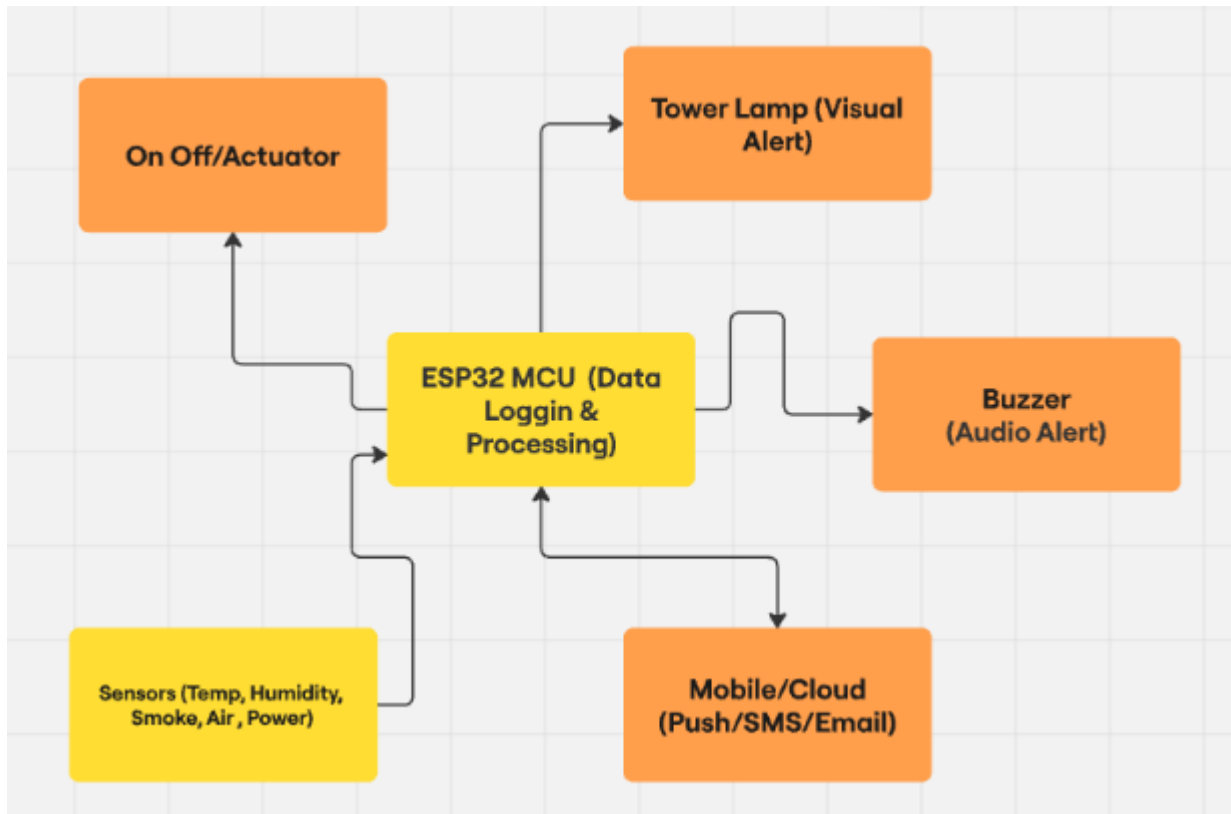


Figure – 1.1 Thesis Organization Flowchart

1.2 Problem Statement

Many conventional data center monitoring and cooling systems face operational limitations that affect efficiency, reliability, and safety. In smaller facilities especially, cooling devices are often controlled manually or operate continuously without considering actual thermal demand. This can lead to unnecessary energy consumption or inadequate cooling during critical temperature increases.

Another major challenge is the lack of integrated environmental monitoring. Some systems focus only on temperature control while ignoring other important parameters such as humidity, smoke, and air quality. Without comprehensive monitoring, abnormal conditions may go undetected until they begin to affect equipment performance or create safety risks.

Remote monitoring and instant alert capabilities are also often absent in traditional systems. In situations where operators are not physically present, delays in detecting environmental problems can increase the risk of equipment failure or service interruption.

These limitations highlight the need for an intelligent system capable of continuously monitoring environmental parameters, automatically controlling cooling equipment, providing multi-layered alarm protection, and enabling remote access for operators. This project addresses these needs through an integrated IoT-based solution.

1.3 Objectives of the Project

The main objective of this project is to develop an intelligent IoT-based system for environmental monitoring and automatic cooling control in a data center environment.

The specific objectives are:

- To monitor temperature, humidity, smoke, air quality, eCO₂, and TVOC in real time.
- To implement automatic dual air-conditioner control using predefined temperature setpoints.
- To provide standby cooling support through a secondary air-conditioning unit for improved reliability.
- To develop a visual, audible, and remote alert mechanism for abnormal environmental conditions.
- To enable remote monitoring and command functionality using a Telegram bot.
- To store control setpoints in non-volatile memory for configuration retention during power interruptions.
- To improve energy efficiency, safety, and operational stability in server room environments.

1.4 Scope of the Project

This project focuses on the design and implementation of an IoT-based environmental monitoring and cooling control system intended for small to medium-sized data centers, server rooms, and network equipment rooms.

The scope includes:

- Real-time monitoring of environmental parameters using multiple sensors.
- Automatic control of two air-conditioning units based on temperature thresholds.
- Manual and remote control of cooling devices.
- Alarm generation using LEDs, buzzer, and tower lamp indications.
- Telegram-based status updates and critical event alerts.
- Local monitoring through an OLED display.

The project primarily emphasizes environmental monitoring, cooling automation, and safety alerts. Advanced features such as cloud analytics, predictive control, and integration with enterprise building management systems are considered beyond the current scope and may be explored in future work.

1.5 Significance of the Study

The significance of this study lies in its practical contribution to improving environmental control and operational reliability in data center applications. By integrating monitoring, cooling control,

alarms, and remote communication into a single system, the proposed solution provides a more unified approach compared with fragmented traditional methods.

This study also contributes to energy efficiency by enabling cooling based on actual environmental demand rather than fixed operating schedules. In addition, the dual AC strategy introduces an added level of redundancy, which can improve system resilience in the event of thermal stress or primary cooling failure.

From a safety perspective, integrating smoke detection, air quality monitoring, and alarm mechanisms improves early detection of abnormal conditions and supports faster response. The use of IoT communication further enhances accessibility by allowing operators to monitor and interact with the system remotely.

Overall, this work demonstrates a low-cost and scalable approach that can be adapted for practical deployment in modern server environments.

1.6 Organization of the Thesis

This thesis is organized into six chapters.

Chapter 1 presents the introduction, including background, problem statement, objectives, scope, and significance of the study.

Chapter 2 discusses background concepts and reviews related work in data center cooling, environmental monitoring, IoT systems, and automated control.

Chapter 3 presents the design and implementation of the proposed system, including hardware architecture, software design, and control logic.

Chapter 4 presents the experimental results, performance analysis, and discussion of system operation.

Chapter 5 provides the conclusion of the study and summarizes the key findings.

Chapter 6 presents recommendations and future scope for improving and extending the proposed system.

Chapter 2 — Background and Literature Review

2.1 Introduction

Data centers have become essential to modern digital infrastructure, supporting cloud services, enterprise applications, communication systems, and large-scale data storage. As data center workloads continue to increase, maintaining reliable environmental conditions has become a major operational requirement. Cooling performance, air quality, and environmental safety directly influence equipment reliability, energy consumption, and service continuity.

This chapter presents the background concepts and related studies relevant to the proposed IoT-based data center cooling and environmental monitoring system. It reviews data center cooling principles, environmental monitoring technologies, IoT-based control systems, smart HVAC approaches, and remote alert mechanisms. It also identifies gaps in existing approaches that motivate the development of the proposed system.

2.2 Data Center Cooling Fundamentals

Cooling is one of the most critical aspects of data center operation. Servers, switches, and storage equipment generate heat continuously, and this heat must be removed efficiently to maintain safe operating temperatures.

Traditional cooling systems commonly rely on room-level air-conditioning units operating at fixed temperature setpoints. While simple to implement, these systems often suffer from inefficient cooling distribution, excessive energy use, and poor adaptability to changing thermal loads.

A major challenge in data center cooling is the formation of localized hot spots. These occur when airflow distribution is uneven or cooling capacity does not match equipment heat generation. Hot spots can increase component stress and reduce system reliability.

Modern cooling strategies increasingly focus on adaptive control, where cooling systems respond dynamically to measured environmental conditions. This approach can improve thermal stability while reducing unnecessary energy consumption.

2.3 Environmental Monitoring in Data Centers

Environmental monitoring is essential for maintaining safe and stable conditions inside server rooms. While temperature is a primary parameter, other environmental factors are also important.

Humidity control helps prevent electrostatic discharge at low humidity levels and condensation-related risks at high humidity levels. Both conditions can negatively affect electronic equipment.

Air quality has also become a growing concern in sensitive environments. Elevated concentrations of contaminants, volatile compounds, or carbon dioxide may affect both equipment integrity and personnel safety.

Smoke detection is particularly important because early warning can help reduce fire-related risks. Traditional fire detection systems often operate separately from environmental monitoring systems, which can delay coordinated response.

Integrated monitoring systems that combine multiple environmental parameters provide a more complete understanding of operating conditions and improve response capability.

2.4 Internet of Things (IoT) in Monitoring and Control

The Internet of Things has transformed conventional monitoring systems by enabling connected devices to collect, process, and exchange data in real time.

An IoT system typically includes sensors, a processing unit, communication interfaces, and a user interaction platform. In monitoring applications, this architecture supports continuous environmental observation and automated decision-making.

Microcontrollers such as the ESP8266 and ESP32 are widely used in IoT systems because they provide built-in wireless communication, flexible interfacing, and low-cost implementation.

In data center applications, IoT makes it possible to combine environmental sensing, automatic cooling control, alarm generation, and remote communication within a single framework. This improves responsiveness, reduces manual intervention, and increases operational visibility.

2.5 Smart HVAC and Automated Cooling Control

Smart HVAC systems use sensor feedback and control logic to regulate cooling equipment more efficiently than conventional fixed-operation systems.

One common approach is rule-based control, where cooling devices are activated or deactivated based on measured values compared with predefined thresholds. This method is practical, reliable, and well suited to embedded systems.

In some applications, dual-unit cooling strategies are used to improve redundancy. A primary cooling unit handles normal control requirements, while a secondary unit acts as a backup during elevated thermal conditions or equipment failure.

Compared with manual cooling operation, automated control offers several advantages:

- Faster response to temperature changes
- Improved cooling efficiency
- Reduced operator dependency
- Better protection against overheating
- Enhanced system reliability through redundancy

These principles support the cooling control approach used in the proposed system.

2.6 Remote Monitoring and Alert Systems

Remote monitoring has become increasingly important, especially in facilities where continuous on-site supervision is not always practical.

Traditional remote alert methods have often relied on SMS notifications or proprietary supervisory systems. More recent approaches use internet-based messaging platforms to provide faster, lower-cost communication.

Messaging-based platforms allow environmental data, alarms, and system status to be transmitted in real time. They may also support remote command functions, allowing operators to interact with the system when necessary.

This project uses Telegram as a remote monitoring and control interface because it supports instant alerts, status messaging, and bidirectional communication. This improves accessibility and operator response capability.

2.7 Related Work and Existing Systems

Several studies have explored sensor-based monitoring and automatic cooling control for server rooms and related environments.

Existing research has demonstrated that IoT-based monitoring can improve visibility into environmental conditions and support more efficient cooling decisions. Other studies have shown that automatic control strategies can reduce energy use compared with fixed cooling schedules.

However, many existing implementations focus only on monitoring without integrating automated control, or they implement temperature control without broader safety functions such as smoke detection and air quality monitoring.

Some systems provide alerts but do not support remote control. Others rely on expensive building management platforms that may not be practical for smaller installations.

These limitations indicate that there remains a need for integrated, low-cost solutions that combine monitoring, control, alarms, and remote access.

2.8 Research Gap and Motivation

Based on the reviewed background and related studies, several gaps can be identified:

- Limited integration of multi-parameter monitoring and cooling control in a single system
- Insufficient use of dual air-conditioner redundancy in low-cost implementations
- Lack of combined environmental monitoring, alarm indication, and remote control
- Limited use of persistent configuration storage for reliable setpoint management

- Incomplete integration of practical messaging platforms for monitoring and command functions

These gaps provide the motivation for the proposed system.

The system developed in this work addresses these limitations by integrating environmental sensing, automatic dual AC control, alarm mechanisms, and remote Telegram-based interaction into a unified IoT platform.

2.9 Summary

This chapter presented the background concepts and literature relevant to the proposed IoT-based data center cooling and environmental monitoring system. It discussed cooling fundamentals, environmental monitoring requirements, IoT technologies, automated HVAC control, remote monitoring approaches, and limitations in existing systems.

The review shows that while significant progress has been made, opportunities remain for integrated, practical, and scalable solutions. These findings provide the foundation for the design and implementation presented in the next chapter.

Chapter 3 — Design and Implementation

3.1 Introduction

This chapter presents the design and implementation of the proposed IoT-based data center cooling, environmental control, and alert data logging system. The system was developed to provide continuous monitoring of critical environmental parameters, automatic cooling control, safety alarm generation, and remote monitoring through an IoT communication interface.

The design combines hardware and software modules into a unified system capable of sensing environmental conditions, processing data, making control decisions, and responding automatically when abnormal conditions are detected. A modular approach was used to improve reliability, simplify implementation, and support future scalability.

Component	QTY	Model	Specification	Function
Microcontroller	1	ESP8266	80 MHz, Wi-Fi Enabled, 3.3V	Main processing and control
Temperature & Humidity Sensor	1	AHT21	$\pm 0.3^{\circ}\text{C}$, $\pm 2\%$ RH	Temperature and humidity monitoring
Air Quality Sensor	1	ENS160	eCO ₂ , TVOC, AQI	Air quality monitoring
Smoke Sensor	1	MQ-2	Analog gas/smoke detection	Smoke detection
OLED Display	1	SSD1306	128×64 I2C	Local status display
Relay Module	1	2-Channel Relay	5V Relay Module	AC1 and AC2 switching
Alarm Indicator	3	Tower Lamp	Multi-color visual alarm	Visual warning
Buzzer	1	Active Buzzer	5V DC	Audible alarm

Table -3.1 Hardware Components and Specifications

3.2 Overall System Architecture

The proposed system consists of five main functional layers:

1. Sensing Layer
2. Processing and Control Layer
3. Actuation Layer
4. Communication Layer

5. Alarm and Indication Layer

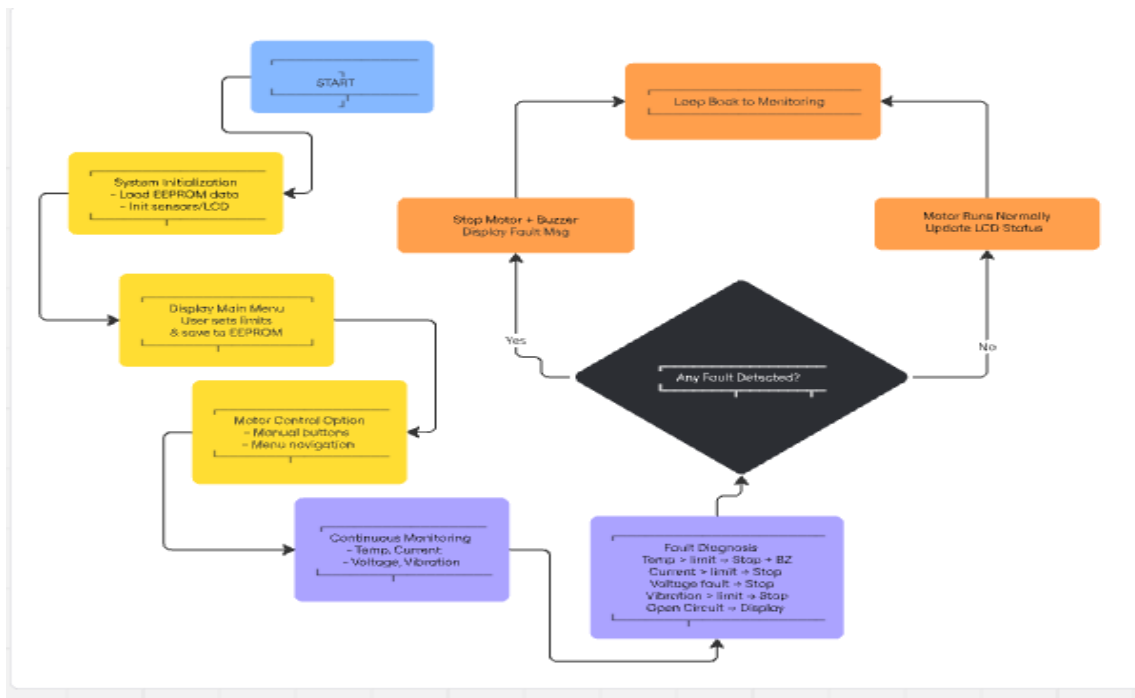


Figure – 3.2 Overall System Architecture Diagram

The sensing layer collects environmental data from multiple sensors installed within the server room. These sensor readings are transmitted to the processing layer, where the controller evaluates conditions and applies control logic.

Based on these decisions, the actuation layer controls the air-conditioning units and alarm devices. At the same time, the communication layer sends data and alerts to remote users through the Telegram platform.

3.3 Hardware Design

Microcontroller Unit

The system is centered around the ESP8266 microcontroller, which acts as the main control unit.

Its functions include:

- Reading environmental sensor data
- Processing control logic
- Switching AC relays
- Managing alarm outputs
- Communicating with the Telegram bot
- Storing temperature setpoints in EEPROM

3.4 Cooling Control Design

The system uses a dual air-conditioner strategy.

Primary Cooling Unit (AC1)

AC1 operates automatically using temperature setpoints.

Control logic:

- If Temperature > High Setpoint → AC1 ON
- If Temperature < Low Setpoint → AC1 OFF

This creates stable temperature regulation while avoiding unnecessary switching.

Standby Cooling Unit (AC2)

AC2 operates as a backup cooling unit.

If:

Temperature > (High Setpoint + 2°C)

Then:

AC2 turns ON automatically.

This adds redundancy and protects against thermal overload.

3.5 Alarm and Safety Design

The system includes multiple alarm mechanisms.

Visual Alarm Indicators

- Temperature alarm LED
- Humidity alarm LED
- Smoke alarm LED
- Tower lamp indication

Audible Alarm

A buzzer activates during critical environmental conditions.

Alarm is triggered when:

- Temperature exceeds limits
- Humidity exceeds limits
- Smoke exceeds threshold

If multiple critical conditions occur simultaneously, the system enters critical alarm mode.

Parameter	Trigger Condition	Alarm Response	System Action
Temperature	Temperature > High Setpoint	Temperature LED ON	AC1 Activated
Critical Temperature	Temperature > High Setpoint + 2°C	Temperature LED ON	AC2 Activated (Backup Cooling)
Low Temperature	Temperature < Low Setpoint	No Alarm	AC1 OFF
Humidity	Humidity outside preset limits	Humidity LED ON	Warning Generated
Smoke	Smoke Value > Threshold	Smoke LED ON	Buzzer + Tower Lamp Activated
Air Quality	AQI exceeds safe level	Warning Alert	Telegram Notification
Multi-Parameter Critical Condition	Temperature + Humidity + Smoke exceed limits	Full Alarm Triggered	Buzzer + Tower Lamp + Telegram Alert

Table -3.5 Alarm Trigger Conditions

3.6 User Interface Design

Local Interface

An SSD1306 OLED display provides local system diagnostics.

Display pages show:

- Temperature and humidity
- Air quality values
- Smoke level
- AC operating status

The display cycles automatically between pages.

Manual Control Interface

Two push buttons provide manual override:

- Button 1 → AC1 control
- Button 2 → AC2 control

Manual override improves flexibility during maintenance or emergency conditions.

3.7 Software Design

The system software was developed using the Arduino IDE.

The software structure includes:

- Sensor reading module
- Control logic module
- Alarm handling module
- EEPROM management module
- Telegram communication module
- Display update module

The main program runs continuously in a loop:

1. Read sensors
2. Process environmental conditions
3. Execute control logic
4. Update alarms
5. Send Telegram updates
6. Update OLED display
7. Receive user commands

This loop repeats throughout system operation.

3.8 Telegram Bot Integration

Remote monitoring and control are implemented using a Telegram bot through Telegram.

Functions include:

- Periodic status messages
- Critical alarm notifications
- Remote status command
- Remote control capability

Example:

/status → Returns current environmental data

Periodic status reports are transmitted automatically every two minutes.

This provides continuous remote visibility of system conditions.

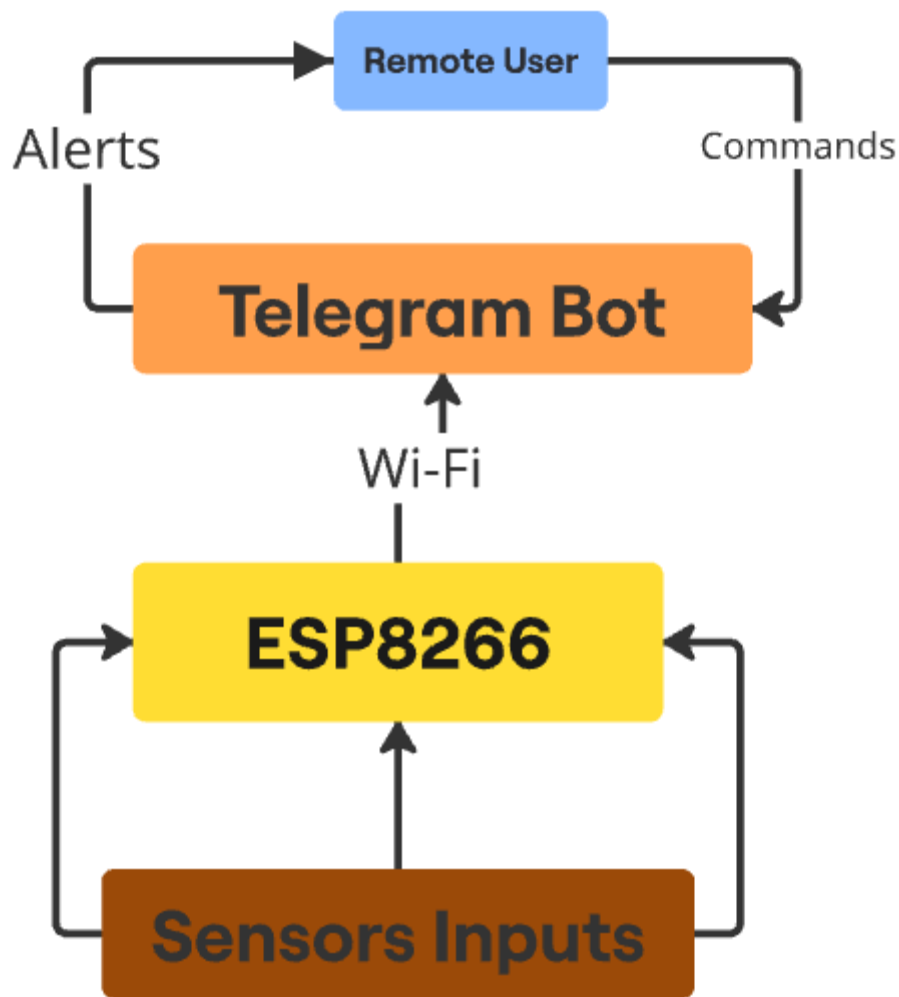


Figure – 3.8 Telegram Communication Workflow

3.9 EEPROM-Based Setpoint Storage

Temperature setpoints are stored in internal EEPROM.

Stored parameters:

- Lower temperature limit
- Upper temperature limit
- Humidity limits
- Smoke threshold

This ensures that configuration values are retained even after power interruptions.

On startup:

- Setpoints are loaded from memory
- System resumes operation automatically

This improves reliability.

3.10 System Implementation

The system was implemented by integrating sensors, control relays, alarms, display, and communication modules into a functional prototype.

Implementation steps included:

- Hardware assembly
- Sensor interfacing
- Relay integration
- Firmware development
- Control logic testing
- Telegram communication setup
- System calibration

After integration, the prototype was tested under simulated environmental conditions to verify monitoring accuracy, cooling response, and alarm functionality.

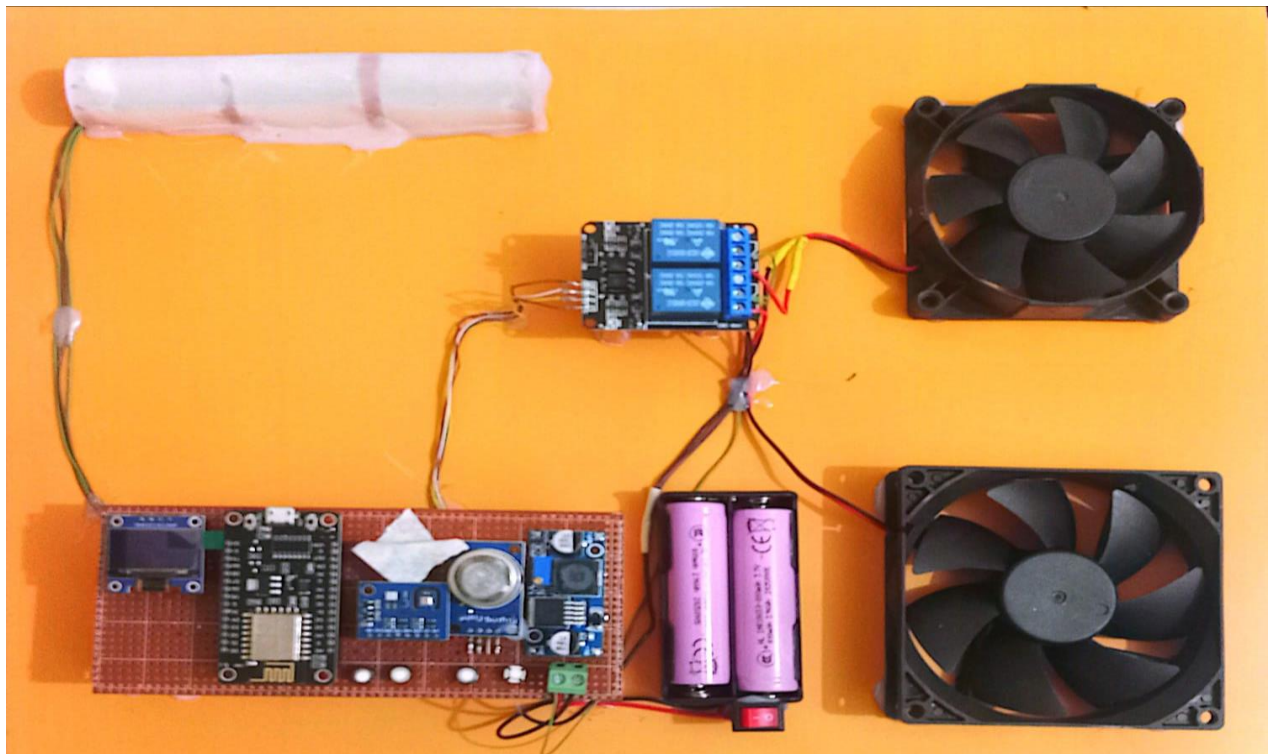


Figure – 3.10 Hardware Prototype Photograph

3.11 Design Considerations

Several practical factors were considered during design:

Reliability

Redundant cooling through AC2 improves system safety.

Scalability

Additional sensors or cooling units can be added later.

Cost Efficiency

Low-cost components were selected while maintaining performance.

Safety

Integrated alarms improve early warning response.

Maintainability

Modular design simplifies troubleshooting and upgrades.

Chapter 4 — Results and Discussion

4.1 Introduction

This chapter presents the results obtained from the implementation and testing of the proposed IoT-based data center cooling, environmental control, and alert data logging system. The purpose of this chapter is to evaluate system performance, analyze operational behavior, and discuss how effectively the developed system meets the project objectives.

The results focus on environmental monitoring accuracy, cooling control performance, alarm effectiveness, remote communication reliability, and overall system stability under different test conditions.

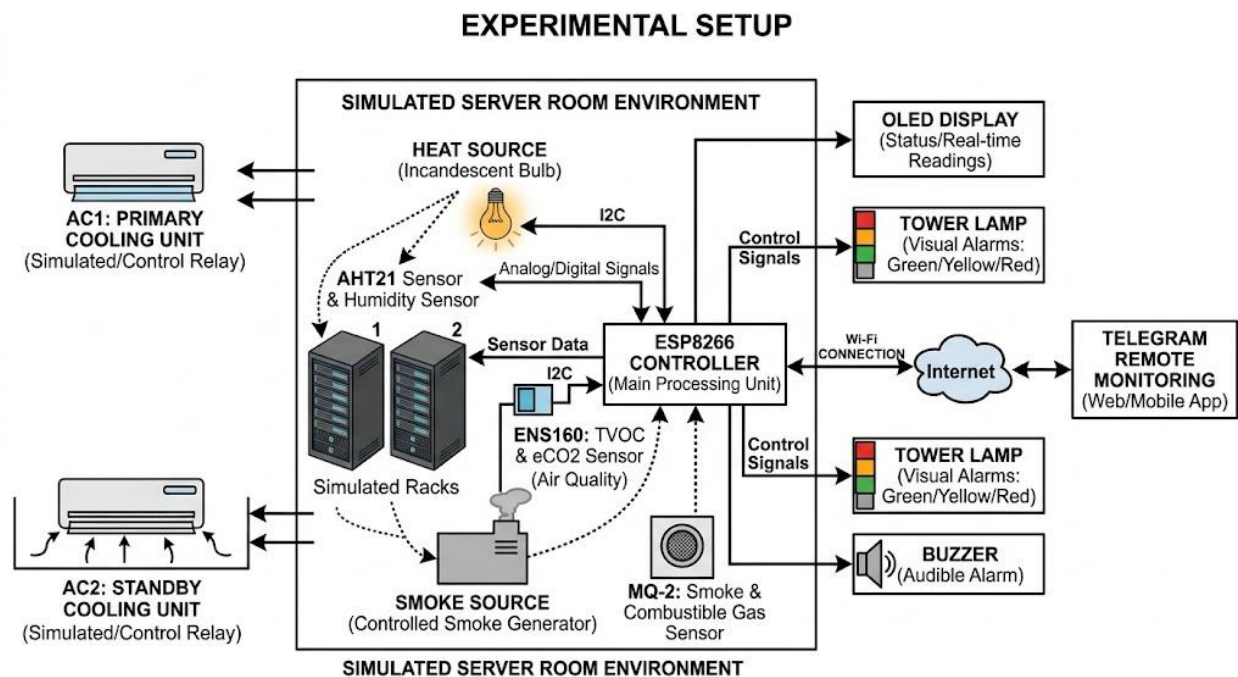


Figure – 4.1 Experimental Setup

4.2 Experimental Setup

The prototype system was tested in a simulated server room environment designed to reproduce practical operating conditions.

The experimental setup included:

- ESP8266-based control unit
- Temperature and humidity sensing using the AHT21
- Air quality monitoring using the ENS160
- Smoke detection using the MQ-2

- Dual air-conditioner relay control
- Alarm indicators, tower lamp, and buzzer
- Telegram-based remote monitoring through Telegram

Multiple test scenarios were conducted to evaluate:

- Normal operating conditions
- Elevated temperature conditions
- Humidity deviations
- Smoke detection response
- Critical multi-parameter alarm conditions

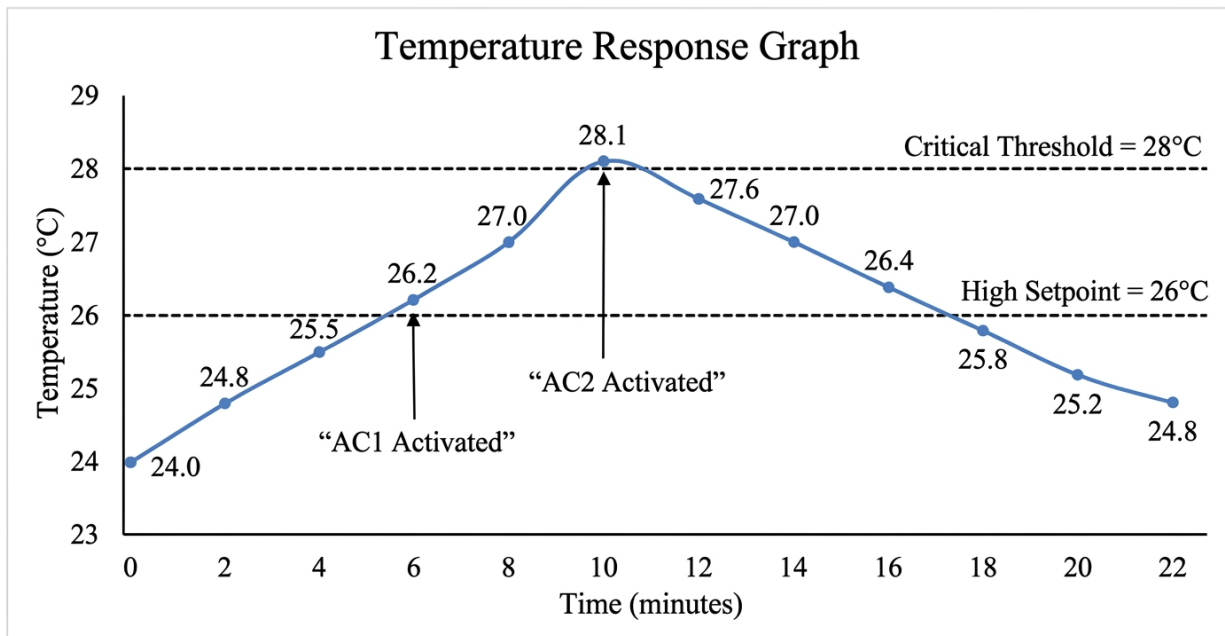


Figure – 4.2 Temperature Response Graph

4.3 Environmental Monitoring Performance

Temperature Monitoring

The temperature monitoring subsystem showed stable and reliable performance throughout testing.

The system successfully detected temperature variations in real time and responded consistently to thermal changes introduced during testing.

When room temperature exceeded the upper setpoint:

- AC1 activated automatically.

When temperature crossed the critical threshold:

- AC2 was engaged as a backup cooling response.

This confirmed proper operation of the dual-stage cooling logic.

Humidity Monitoring

Humidity monitoring performance was also stable.

The system detected abnormal humidity conditions and triggered corresponding alerts when predefined thresholds were exceeded.

The humidity alarm indicators operated as intended, and warning conditions were transmitted successfully through Telegram notifications.

Air Quality and Smoke Detection

The air quality subsystem provided consistent monitoring of:

- eCO₂
- TVOC
- AQI

The system successfully detected degraded air quality conditions during testing and triggered alarms when thresholds were exceeded.

Smoke detection using the MQ-2 sensor also responded reliably.

During smoke test simulation:

- Smoke alarm LED activated
- Tower lamp activated
- Buzzer activated
- Telegram alert transmitted immediately

These results confirm effective early warning capability.

Alarm Response Diagram

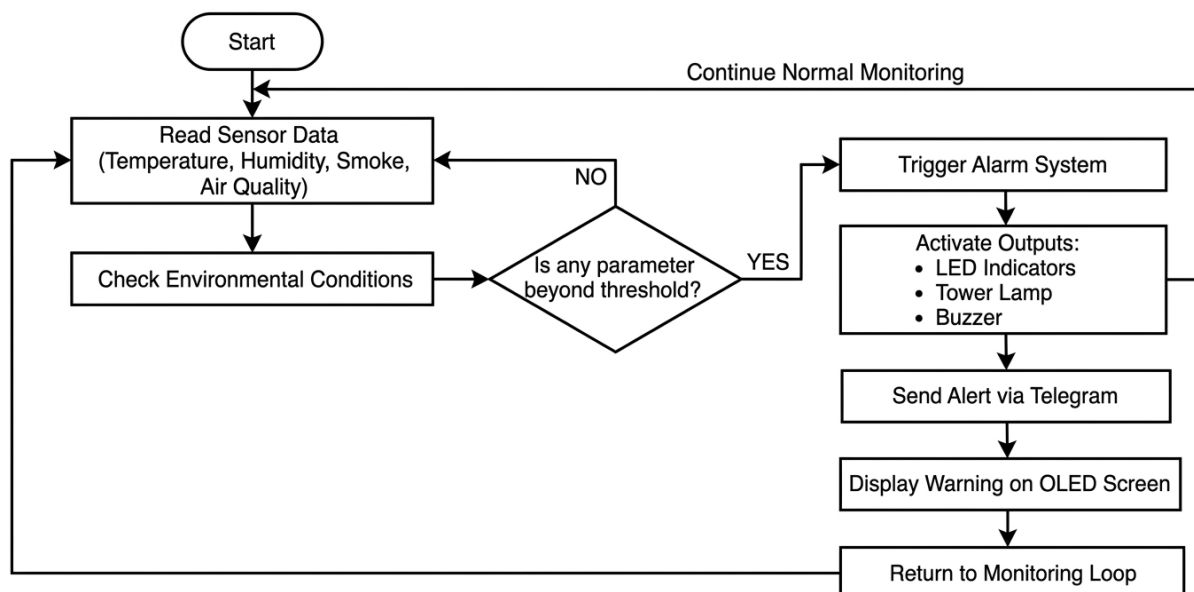


Figure – 4.3 Alarm Response Diagram

Feature/Criteria	Conventional Monitoring Systems	Proposed IoT-Based System
Monitoring Method	Often manual or human-initiated sampling	Real-time multi-parameter (Temp, Humidity, Air Quality, Smoke)
Cooling Strategy	Fixed operating schedules or manual control	Automatic adaptive control based on thermal demand
Redundancy	Typically lacks secondary failsafe stages	Dual AC redundancy (AC1 Primary + AC2 Standby)
Alarm Mechanisms	Limited to local on-site indicators (buzzers/lights)	Integrated multi-level alarms (Tower lamp, buzzer, and remote Telegram)
Remote Access	Minimal or non-existent remote interaction	Full remote monitoring and control via Telegram bot
Automation Level	Lower; requires physical technician intervention	High ; reduces the need for constant on-site supervision
Configuration Storage	Parameters may be lost during power failure	Persistent storage (EEPROM-based setpoint retention)
Implementation Cost	Expensive industrial/proprietary solutions	Low-cost and scalable using open-source hardware (ESP8266)

Table - 4.3 Comparative System Evaluation Comparative System Evaluation

4.4 Cooling Control Performance

The automatic cooling control mechanism performed as designed.

The primary cooling unit responded to temperature changes using the defined high and low setpoints.

The hysteresis control approach reduced frequent relay switching and improved control stability.

The standby cooling unit provided an additional safety layer by activating only during elevated thermal conditions.

Key observations:

- Stable temperature regulation achieved
- Reduced unnecessary switching observed
- Backup cooling activated correctly
- Thermal control remained within safe operating limits

Compared with fixed-schedule cooling methods, the proposed system showed improved responsiveness and more adaptive operation.

4.5 Alarm System Evaluation

The integrated alarm system performed reliably under all tested abnormal conditions.

Visual alarms correctly indicated:

- Temperature alarm conditions
- Humidity alarm conditions
- Smoke detection events

The audible buzzer provided immediate warning during critical events.

The tower lamp also improved condition visibility.

In multi-parameter fault scenarios, the system entered critical alarm mode and activated all alarm mechanisms simultaneously.

This multi-layer alarm strategy improved response capability and safety performance.

4.6 Remote Monitoring and Communication Performance

The Telegram communication system performed reliably during testing.

Periodic status reports were delivered successfully at scheduled intervals.

Critical event notifications were transmitted without significant delay.

Remote status requests also operated correctly.

Example functions tested:

- Status command response
- Periodic telemetry updates
- Alarm notifications
- Remote monitoring access

The use of Telegram provided a practical and low-cost communication interface compared with traditional alert systems.

4.7 System Reliability and Stability

The system operated continuously during the test period without major instability.

Several reliability features contributed to performance:

- EEPROM setpoint retention after restart
- Stable sensor data acquisition
- Reliable relay operation
- Consistent Wi-Fi communication
- Functional alarm behavior

No significant data loss or critical control failures were observed during testing.

These results indicate acceptable operational stability for the prototype system.

4.8 Comparative Discussion

Compared with conventional environmental monitoring systems, the proposed system offers several advantages:

Conventional Systems

- Manual monitoring
- Fixed cooling schedules
- Limited alarms
- Minimal remote access
- Lower automation

Proposed System

- Real-time multi-parameter monitoring
- Automatic adaptive cooling control
- Dual AC redundancy
- Integrated alarms
- Remote monitoring and control
- Lower implementation cost

This comparison demonstrates practical benefits in efficiency, safety, and reliability.

4.9 Limitations

Although the system performed effectively, several limitations were observed.

- Testing was conducted on a prototype-scale environment.
- Long-term sensor drift was not extensively evaluated.
- Remote communication depends on network availability.
- The system currently supports only two air-conditioning units.

These limitations provide direction for future improvement.

4.10 Discussion of Findings

The results indicate that the developed system successfully meets the main project objectives.

The system was able to:

- Monitor environmental conditions in real time
- Control cooling devices automatically
- Trigger reliable alarms
- Provide remote access through IoT communication
- Improve operational safety through redundancy

From a practical engineering perspective, the dual AC strategy and integrated alarm design significantly strengthen the reliability of the proposed solution.

Chapter 5 — Conclusion

5.1 Introduction

This chapter presents the conclusion of the study based on the design, implementation, and evaluation of the proposed IoT-based data center cooling, environmental control, and alert data logging system. The chapter summarizes the major outcomes of the project, highlights its contributions, discusses limitations, and presents recommendations for future improvement.

The objective of this project was to develop a practical and intelligent system capable of monitoring critical environmental parameters, controlling cooling equipment automatically, generating alarms during abnormal conditions, and enabling remote monitoring through IoT communication. Based on the experimental results presented in the previous chapter, the developed system achieved these objectives successfully.

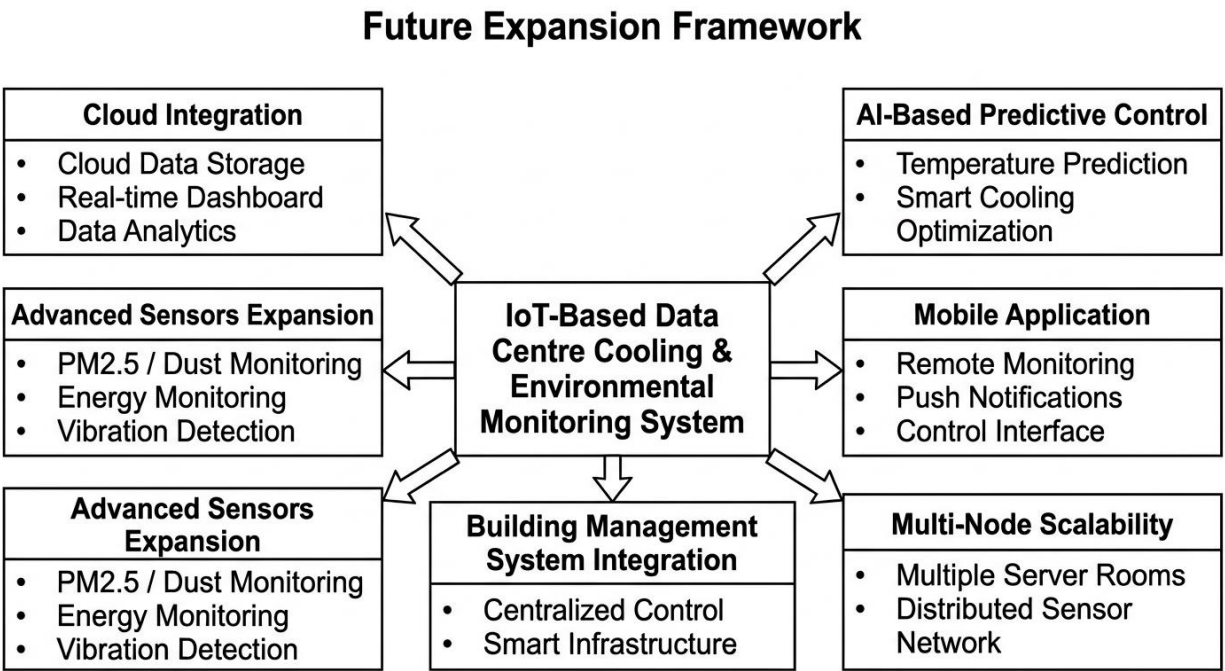


Figure – 5.1 Future Expansion Framework

Contribution Area	Technical Description	Operational Benefit
Integrated Monitoring	Unified framework for Temperature, Humidity, Smoke, and Air Quality (AQI, eCO2, TVOC)	Eliminates the need for fragmented, independent monitoring tools
Dual-Stage Cooling	Automated logic for Primary (AC1) and Standby Failsafe (AC2) cooling units	Ensures high availability and prevents thermal runaway during primary failures
Reliability Features	EEPROM-based storage for thermal setpoints and configuration parameters	Prevents data/setting loss and ensures autonomous resumption after power outages
Multi-Tiered Alerting	Integration of local visual/audible alarms with instant remote Telegram notifications	Facilitates immediate awareness for both on-site technicians and remote administrators
Remote Interactivity	Bidirectional communication via a Telegram Bot for status polling and manual control	Reduces the frequency of routine physical interventions in the data hall
Energy Efficiency	Demand-based cooling activation using precise thermal setpoints rather than fixed timers	Minimises unnecessary power consumption by aligning cooling with actual thermal load
System Scalability	Developed using low-cost, open-source hardware (ESP8266/ESP32) and modular logic	Provides a cost-effective alternative to expensive industrial PLC-based solutions

Table 5.1: Summary of Contributions of the Study

5.2 Conclusion of the Study

This project successfully developed and evaluated an IoT-based environmental monitoring and cooling control system designed for data center and server room applications.

The system demonstrated the ability to continuously monitor key environmental parameters, including temperature, humidity, air quality, and smoke conditions. Through integration of multiple sensors, the system provided real-time situational awareness and improved visibility of operating conditions.

The automatic dual air-conditioner control strategy proved effective in maintaining thermal stability. The primary cooling unit responded to temperature conditions based on predefined setpoints, while the standby cooling unit provided an additional layer of protection during elevated thermal conditions. This dual-stage control approach improved system reliability and reduced the risk of overheating.

The integrated alarm mechanisms also performed effectively. Visual indicators, tower lamp signaling, audible alarms, and Telegram notifications provided multiple levels of warning during

abnormal environmental events. This multi-layered alarm approach improved response readiness and strengthened operational safety.

Remote monitoring through Telegram further enhanced system accessibility by enabling users to receive status updates and alerts without requiring constant physical presence in the server room.

Overall, the developed system demonstrated that combining IoT monitoring, automatic control, alarm protection, and remote communication in a unified platform provides a practical and scalable solution for improving environmental management in data centers.

Improvement Area	Technical Description	Expected Impact
Predictive Analytics (AI)	Implementing Machine Learning models to analyse thermal trends ($\Delta T/\Delta t$) and predict "hot spots" before they breach setpoints	Shifts system from reactive to proactive cooling, significantly reducing thermal stress on hardware
Cloud-Based Data Logging	Moving beyond Telegram logs to dedicated platforms like Thingspeak, Firebase, or InfluxDB for long-term historical data storage	Enables comprehensive auditing, performance charting, and SLA compliance reporting
Network Redundancy	Integrating GSM modems or LoRa (Long Range) modules as a fail-safe communication layer	Ensures critical alerts (smoke/fire) are delivered even during Wi-Fi or local ISP outages
Augmented Reality (AR)	Developing a mobile AR interface to overlay real-time sensor data directly onto physical server racks using the device camera	Enhances technician situational awareness and reduces diagnostic time during maintenance
Fine-Grained Power Profiling	Adding inline AC metering (e.g., via PZEM-004T) to track total rack energy consumption and Power Usage Effectiveness (PUE)	Provides a granular view of cooling efficiency versus computational load
Expanded Sensor Suite	Incorporating vibration sensors for cooling fan health and PM2.5 sensors for dust/particulate monitoring	Protects servers from mechanical wear and internal airflow blockages caused by dust accumulation
Industrial SCADA Integration	Adapting the communication layer to support standard industrial protocols like Modbus TCP or BACnet	Facilitates seamless integration into enterprise Building Management Systems (BMS).

Table - 5.2 Proposed Future Improvements

5.3 Contributions of the Study

This study makes several practical and technical contributions.

Integrated Multi-Parameter Monitoring

The system combines temperature, humidity, air quality, and smoke detection into a unified monitoring framework.

Automated Cooling Control

The project implemented a dual air-conditioner control strategy that improves cooling reliability and introduces redundancy.

Alarm and Safety Integration

The combination of visual alarms, audible alarms, and remote notifications improves abnormal condition detection and response capability.

IoT-Based Remote Monitoring

Integration with Telegram provides a low-cost and practical remote communication interface for system monitoring and alerts.

Persistent Configuration Storage

EEPROM-based storage improves reliability by retaining control setpoints after power interruptions.

These contributions support both practical deployment and future development opportunities.

5.4 Limitations of the Study

Although the developed system performed effectively, several limitations should be acknowledged.

- The evaluation was conducted using a prototype-scale setup rather than a full-scale data center environment.
- Long-term performance analysis, including sensor drift and extended durability testing, was limited.
- Remote communication performance depends on stable internet connectivity.
- The current implementation supports only two air-conditioning units without additional expansion hardware.

These limitations do not reduce the effectiveness of the proposed system but identify areas for future improvement.

5.5 Recommendations and Future Improvements

Based on the results and observed limitations, several future enhancements are recommended.

Cloud-Based Data Logging

Future versions of the system could integrate cloud platforms for long-term data storage, analytics, and visualization.

AI-Based Predictive Cooling

Predictive algorithms could improve cooling efficiency by anticipating thermal changes before threshold violations occur.

Expanded Sensor Network

Additional sensing parameters such as particulate matter, vibration, and electrical load monitoring could further improve environmental awareness.

Multi-Unit Cooling Scalability

The system could be expanded to support larger data center installations with multiple cooling units.

Building Management Integration

Future integration with building management systems could support centralized monitoring and control.

These improvements could enhance scalability, intelligence, and long-term operational performance.

5.6 Final Remarks

Modern data centers require reliable environmental control to protect critical infrastructure and maintain continuous operation. This study demonstrates that a low-cost IoT-based approach can provide effective monitoring, automated cooling control, alarm protection, and remote accessibility within a single integrated system.

From a practical engineering perspective, the proposed system offers a strong foundation for scalable and intelligent data center environmental management.

5.7 Summary

This chapter presented the conclusion of the study, summarized the major findings, identified the contributions and limitations of the project, and outlined recommendations for future work.

The results confirm that the proposed IoT-based data center cooling and environmental monitoring system achieved its intended objectives and provides a practical solution for improving safety, reliability, and efficiency in modern server environments.

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Appendix

Program:

```
/******  
SMART IoT-BASED DATA CENTER COOLING & ENVIRONMENT MONITORING  
MCU: ESP8266  
Sensors: ENS160 + AHT21, MQ-2  
Display: OLED SSD1306  
Control: Dual AC (Primary + Standby)  
*****/  
  
#include <ESP8266WiFi.h>  
#include <WiFiManager.h>  
#include <Wire.h>  
#include <EEPROM.h>  
  
#include <Adafruit_GFX.h>  
#include <Adafruit_SSD1306.h>  
  
#include <WiFiClientSecure.h>  
#include <UniversalTelegramBot.h>  
  
#include <Adafruit_AHTX0.h>  
#include "ScioSense_ENS160.h"  
  
/* ===== OLED ===== */  
#define SCREEN_WIDTH 128  
#define SCREEN_HEIGHT 64  
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);  
  
/* ===== SENSOR ===== */  
Adafruit_AHTX0 aht;  
ScioSense_ENS160 ens160(ENS160_I2CADDR_1); // 0x53 ENS160+AHT21  
  
/* ===== PINS ===== */  
#define MQ2_PIN A0  
  
#define LED_TEMP D5  
#define LED_HUM D6  
#define LED_SMOKE D7  
#define BUZZER D8  
  
#define AC1_RELAY D0 // PRIMARY AUTO  
#define AC2_RELAY D4 // STANDBY
```

```

#define BTN_AC1      D3
#define BTN_AC2      D9    // change if conflict

/* ===== TELEGRAM ===== */
#define BOT_TOKEN    "8143345279:AAHEKn1yBwDFiEHp1s8mDgGqb1f-rX7xuT0"
#define CHAT_ID      "5668796260"

WiFiClientSecure client;
UniversalTelegramBot bot(BOT_TOKEN, client);

/* ===== DATA ===== */
float temperature, humidity;
uint16_t eco2, tvoc;
uint8_t aqi;
int smokeValue;

/* ===== SETPOINT STRUCT ===== */
struct SETPOINT {
    float tLow, tHigh;
    float hLow, hHigh;
    int smoke;
} sp;

/* ===== TIMERS ===== */
unsigned long lastTelegram = 0;
unsigned long lastDisplay = 0;
unsigned long lastButton = 0;
int displayPage = 0;

/* ===== FUNCTION DECLARATION ===== */
void initSensors();
void readSensors();
void controlLogic();
void updateDisplay();
void handleTelegram();
void loadEEPROM();
void saveEEPROM();

/* ===== SETUP ===== */
void setup() {
    Serial.begin(9600);
    EEPROM.begin(64);

```

```

pinMode(LED_TEMP, OUTPUT);
pinMode(LED_HUM, OUTPUT);
pinMode(LED_SMOKE, OUTPUT);
pinMode(BUZZER, OUTPUT);

pinMode(AC1_RELAY, OUTPUT);
pinMode(AC2_RELAY, OUTPUT);

pinMode(BTN_AC1, INPUT_PULLUP);
pinMode(BTN_AC2, INPUT_PULLUP);

digitalWrite(AC1_RELAY, HIGH);
digitalWrite(AC2_RELAY, HIGH);

WiFiManager wm;
wm.autoConnect("DataCenter-IoT");

client.setInsecure();
bot.sendMessage(CHAT_ID, "✔ Data Center System ONLINE", "");

Wire.begin();
display.begin(SSD1306_SWITCHCAPVCC, 0x3C);

initSensors();
loadEEPROM();
}

/* ===== LOOP ===== */
void loop() {
    readSensors();
    controlLogic();
    handleTelegram();

    if (millis() - lastDisplay > 3000) {
        updateDisplay();
        lastDisplay = millis();
    }

    if (millis() - lastTelegram > 120000) {
        String msg =
            "📊 Data Center Status\n"
            "Temp: " + String(temperature) + " C\n" +
            "Hum: " + String(humidity) + " %\n" +
            "eCO2: " + String(eco2) + " ppm\n" +

```

```

        "TVOC: " + String(tvoc) + " ppb\n" +
        "Smoke: " + String(smokeValue);
    bot.sendMessage(CHAT_ID, msg, "");
    lastTelegram = millis();
}
}

/* ===== SENSOR INIT ===== */
void initSensors() {
    if (!aht.begin()) {
        Serial.println("AHT21 ERROR");
        while (1);
    }

    if (!ens160.begin()) {
        Serial.println("ENS160 ERROR");
        while (1);
    }

    ens160.setMode(ENS160_OPMODE_STD);
}

/* ===== SENSOR READ ===== */
void readSensors() {
    sensors_event_t humEvent, tempEvent;
    aht.getEvent(&humEvent, &tempEvent);

    temperature = tempEvent.temperature;
    humidity = humEvent.relative_humidity;

    if (ens160.available()) {
        ens160.set_envdata(temperature, humidity);
        ens160.measure(true);
        ens160.measureRaw(true);

        eco2 = ens160.geteCO2();
        tvoc = ens160.getTVOC();
        aqi = ens160.getAQI();
    }

    smokeValue = analogRead(MQ2_PIN);
}

/* ===== CONTROL LOGIC ===== */

```

```

void controlLogic() {

    bool tempAlarm = (temperature < sp.tLow || temperature > sp.tHigh);
    bool humAlarm = (humidity < sp.hLow || humidity > sp.hHigh);
    bool smokeAlarm = (smokeValue > sp.smoke);

    digitalWrite(LED_TEMP, tempAlarm);
    digitalWrite(LED_HUM, humAlarm);
    digitalWrite(LED_SMOKE, smokeAlarm);

    // PRIMARY AC AUTO CONTROL
    if (temperature > sp.tHigh) {
        digitalWrite(AC1_RELAY, LOW);
    } else if (temperature < sp.tLow) {
        digitalWrite(AC1_RELAY, HIGH);
    }

    // STANDBY AC (FAILSAFE)
    if (temperature > sp.tHigh + 2) {
        digitalWrite(AC2_RELAY, LOW);
    }

    // ALARM CONDITION
    if (tempAlarm && humAlarm && smokeAlarm) {
        digitalWrite(BUZZER, HIGH);
        bot.sendMessage(CHAT_ID, "🚨 CRITICAL ENVIRONMENT ALARM!", "");
    } else {
        digitalWrite(BUZZER, LOW);
    }

    // MANUAL BUTTON TOGGLE
    if (millis() - lastButton > 300) {
        if (!digitalRead(BTN_AC1)) {
            digitalWrite(AC1_RELAY, !digitalRead(AC1_RELAY));
            lastButton = millis();
        }
        if (!digitalRead(BTN_AC2)) {
            digitalWrite(AC2_RELAY, !digitalRead(AC2_RELAY));
            lastButton = millis();
        }
    }
}

/* ===== OLED DISPLAY ===== */

```

```

void updateDisplay() {
    display.clearDisplay();
    display.setTextSize(1);
    display.setCursor(0, 0);

    if (displayPage == 0) {
        display.println("Data Center Monitor");
        display.print("Temp: "); display.print(temperature); display.println(" C");
        display.print("Hum : "); display.print(humidity); display.println(" %");
    }
    else if (displayPage == 1) {
        display.println("Air Quality");
        display.print("eCO2: "); display.print(eco2); display.println(" ppm");
        display.print("TVOC: "); display.print(tvoc); display.println(" ppb");
    }
    else {
        display.println("Smoke & AC");
        display.print("Smoke: "); display.println(smokeValue);
        display.print("AC1: "); display.println(digitalRead(AC1_RELAY) ? "OFF" :
"ON");
        display.print("AC2: "); display.println(digitalRead(AC2_RELAY) ? "OFF" :
"ON");
    }

    display.display();
    displayPage = (displayPage + 1) % 3;
}

/* ===== EEPROM ===== */
void loadEEPROM() {
    EEPROM.get(0, sp);
    if (isnan(sp.tLow)) {
        sp = {22, 26, 45, 60, 400};
        saveEEPROM();
    }
}

void saveEEPROM() {
    EEPROM.put(0, sp);
    EEPROM.commit();
}

/* ===== TELEGRAM ===== */
void handleTelegram() {

```

```
int n = bot.getUpdates(bot.last_message_received + 1);
for (int i = 0; i < n; i++) {
    String text = bot.messages[i].text;

    if (text == "/status") {
        bot.sendMessage(CHAT_ID, "Temp: " + String(temperature), "");
    }
}
}
```